

Advanced Time Series Analysis

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This presentation provides an analysis of a time series data of gross domestic product (GDP) of the Norway, based on the application of R software. An economic time series model with macroeconomic variables is conducted. The forecast of GDP is performed for further economic decision-making. The components of GDP such as *consumption (CS)*, *export (EX)*, and *gross fixed capital formation (IN)* are added for the multivariate time series analysis.

Time series analysis determines data patterns and trends as well as explains data modeling and forecasting. Since a time series of GDP is non-stationary so some statistical tests are followed to transform the time series of GDP became stationary and integrated of order I.

- Box-Jenkins procedure is used to determine ARMA.
- OLS is used to estimate the models parameters.
- ARIMA models are verified basis of classical statistical test, and Vector Error Correcting Model (VECM) for forecasting GDP.

Data Source

Eurostat:<http://ec.europa.eu/eurostat/data/database>

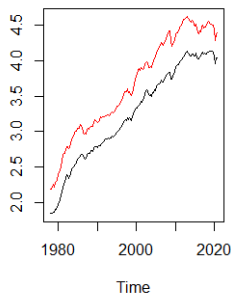
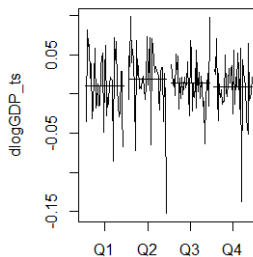
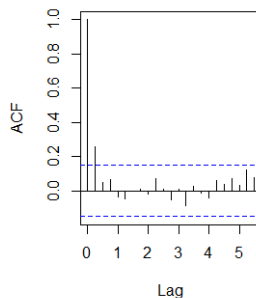
Data for analysis: NorwayData.txt

The GDP is equals to the total expenditure for all goods and services produced within the country in a specific period of time.

$$GDP = C + I + G + (X - M)$$

Times series are analysed in order to understand the nature of fundamental structure and operation of the function.

Analysing GDP of the Norway in billion US\$ in period 1978-2020 in quarter and forecasting GDP for next 12 quarters were performed.

logGDP and logCS**Figure:** Time Series**Figure:** Seasonal plot**Series dlogGDP_ts****Figure:** Autocorrelation

	Std. Deviation	Mean	Maximum	Minimum	Median
GDP	29.668	48.554	101.470	8.886	38.175
CS	18.721	31.637	62.849	6.318	26.056

Table: Descriptive Statistics

Linear Regression:

$$\log(GDP_{ts}) = \beta_0 + \beta_1 TREND + \beta_2 Q_1 + \beta_3 Q_2 + \beta_4 Q_3 + \beta_5 Q_4 + \epsilon,$$

Series dslogGDP_ts

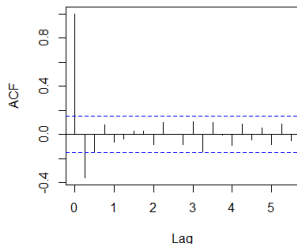


Figure: ACF of log(GDP)

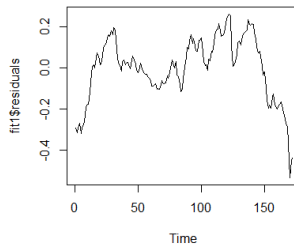


Figure: Residual plot

Multiple R-squared	$R^2 = 0.9529$	95.29% of variance of log(GDP) is explained by regression.
F-statistics	$p\text{-value} = 0.00 < 5\%$	Reject H_0 and the regressors are jointly significant.
p-value	$p\text{-value} = 0.00 < 5\%$	Reject H_0 and Trend is significant.
Ceteris paribus	Estimate Trend =0.0138	Every quarter of GDP increase with 1.38% on average.

Table: Linear regression summary

Linear Regression Equation:

$$\log(GDP_{ts}) = \beta_0 + \beta_1 \log(CS_{ts}) + \beta_2 \log(EX_{ts}) + \beta_3 \log(IN_{ts}) + \epsilon,$$

$$\log(GDP_{ts}) = 0.651 + 0.541 \log(CS_{ts}) + 0.357 \log(EX_{ts}) + 0.121 \log(IN_{ts}) + \epsilon,$$

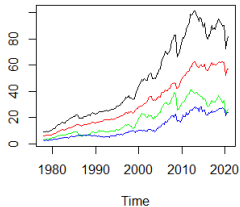


Figure: Multiple graph

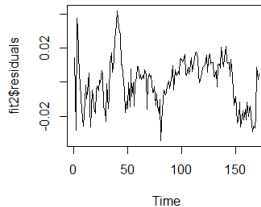


Figure: Time series

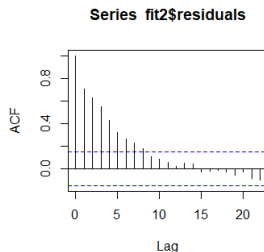


Figure: ACF

The residuals show autocorrelations, and there are not white noise as well, so the model cannot be validated.

Unit root test to check stationary			
$\log(\text{GDP}_{ts})$	$p\text{-value} = 0.916 > 5\%$	Not Reject H_0	Non-stationary
$\log(\text{CS}_{ts})$	$p\text{-value} = 0.972 > 5\%$	Not Reject H_0	Non-stationary
$\text{dlog}(\text{GDP}_{ts})$	$p\text{-value} = 0.000 < 5\%$	Reject H_0	Stationary
$\text{dlog}(\text{CS}_{ts})$	$p\text{-value} = 0.000 < 5\%$	Reject H_0	Stationary
Ljung-Box test for white noise			
$\text{dlog}(\text{GDP}_{ts})$	$p\text{-value} = 0.238 > 5\%$	X-squared = 16.205	White noise
$\text{dlog}(\text{CS}_{ts})$	$p\text{-value} = 0.298 > 5\%$	X-squared = 15.151	White noise

Table: Unit root test and Ljung-Box test

Summary Statistics							
Model	Sigma ²	Ljung-Box test			Log likelihood	AIC	Conclusion
		X-squared	DF	p-value			
ARIMA(3,1,0)	0.0013	7.737	13	0.860	323.38	-638.77	The model is validated.
ARIMA(3,1,1)	0.0013	6.815	13	0.914	323.47	-636.95	The model is validated.

Table: Comparing different ARIMA models

Table: Granger Causality using ADLM(1)

Model 1 = $d\log GDP.0 \sim d\log GDP.1 + d\log CS.1$						
Model 2 = $d\log GDP.0 \sim d\log GDP.1$						
	Res.DF	RSS	Df	Sum of Sq	F	Pr(>F)
1	167	0.21466				
2	168	0.21761	-1	-0.00295	2.295	0.132

Since $p\text{-value} = 0.132 > 5\%$, we do not reject H_0 of no Granger Causality, thus we concluded that $\log(CS)$ has not provide incremental predictive power in forecasting $\log(GDP)$.

Testing for cointegration

1. Engle-Granger ADF

$ADF(1) = -2.613 > -3.41$ (critical value), and $p\text{-value} = 0.0925 > 5\%$, it concludes that $\log(GDP)$ and $\log(CS)$ are no cointegrated, so the further test as Error Correction Model (ECM) test will not be applied.

2. Johansen's approach

Table: Johansen test statistic

	Johansen's trace test				Johansen's maximum eigenvalue			
	test	10pct	5 pct	1pct	test	10pct	5pct	1pct
$r \leq 1$	6.60	7.52	9.24	12.97	6.60	7.52	9.24	12.97
$r=0$	38.85	17.85	19.96	24.60	32.25	13.75	15.67	20.20

Based on Johansen's trace test, for $r = 1$, test statistic is smaller than critical value ($6.60 < 9.24$). For $r = 0$, test statistic is larger than critical value ($38.85 > 19.96$). Thus, $\log(\text{GDP})$ and $\log(\text{CS})$ are cointegrated.

Let δ_t be a stationary time series, the cointegration equation is :

$$\delta_t = 0.053 + \log(\text{GDP}_t) - 1.11 \log(\text{CS}_t)$$

Forecast the value of $\log(\text{GDP})$ for the next 12 quarters by approach:

Figure: Model ARIMA(3,1,0)

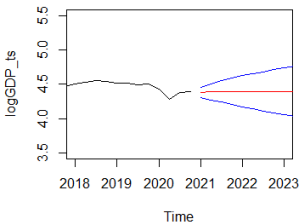


Figure: Model ARIMA(3,1,1)

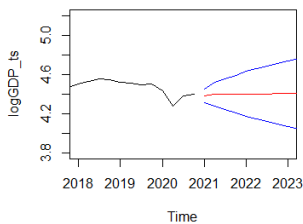
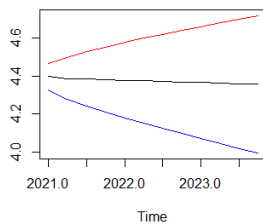


Figure: Model VECM



An expanding-window approach for forecast GDP

	Mean Absolute Error (MAE)	Mean Squared Error (MSE)
ARIMA(3,1,0)	0.02848	0.00183
ARIMA(3,1,1)	0.02878	0.00186
A Diebold-Mariano test	p-value= 0.1085 > 5%	p-value = 0.0404 < 5%
Conclusion	Not significantly different	Significantly different

1. Conclusion

For testing stationary, unit root tests and linear regression test were conducted. Moreover, Engle-Granger ADF tested cointegration, and concluded that there is no cointegration between $\log(\text{GDP})$ and $\log(\text{CS})$. However, based on the result of Johansen's approach, $\log(\text{GDP})$ and $\log(\text{CS})$ are cointegrated.

Moreover, ARIMA(3,1,0) and ARIMA(3,1,1) models, Vector Error Correction model, and an expanding-window approach were followed to forecast the time series of GDP in next 12 quarters. Thus, a Diebold-Mariano test shows that the forecast performance of the two model ARIMA(3,1,0) and ARIMA(3,1,1), one with absolute value loss (MAE), is not significantly different, whereas another one with squared value loss (MSE), is significantly different.

2. Bibliography



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